

Nathan D. DeLay

Department of Agricultural and Resource Economics
Colorado State University
202, Nutrien Agricultural Sciences Building
Fort Collins, CO 80523

Updated: Jan. 15, 2025
Citizenship: United States of America
nathan.delay@colostate.edu

Appointments	Assistant Professor, 2022-current, Colorado State University. Lecturer, 2022-2024, Purdue University. Assistant Professor, 2018-2022, Purdue University.
Education	Ph.D., Economics, May 2018, Washington State University. Fields: Agricultural Economics, Environmental and Natural Resource Economics. Dissertation: “Agricultural Risk Management Policy and Producer Decision Making.” Advisors: Dr. Hayley H. Chouinard, Dr. Thomas L. Marsh M.A., Economics, December 2013, University of Colorado Denver. Master’s Research Project: “The Effect of Crop Insurance Subsidies on Agricultural Productivity.” Advisor: Dr. Brian Duncan B.S., Business Administration and Economics, May 2009, Rocky Mountain College. Magna cum laude
Research Areas	Production Agriculture, Agricultural Policy, Applied Econometrics
Peer Reviewed Work	* graduate student Raynor, Edward J., Anna M. Shadbolt, Melissa K. Johnston, David Augustine, Justin Derner, John P. Ritten, Nathan D Delay, Pedro H. Carvalho, Juan Vargas, Sara E. Place, Kim R. Stackhouse-Lawson. 2025. “Incorporating Virtual Fencing to Manage Yearling Steers on Extensive Rangelands: Spatial Behavior, Growth Performance, and Enteric Methane Emissions.” <i>Frontiers in Veterinary Science</i> 12 - 2025. https://doi.org/10.3389/fvets.2025.1637190 DeLay, Nathan D. and Cory G. Walters. 2025. “Mapping Agents: A Spatial Economic Analysis of Agent Location in the Federal Crop Insurance Program.” <i>American Journal of Agricultural Economics</i> 107(4): 1152–1182. https://doi.org/10.1111/ajae.12503 McKee, Sophie, Nathan D. DeLay, Daniel Mooney, and Stephanie A. Shwiff. 2025. “Externalities in Wild Pig Damages on Crop and Livestock Farms: The Role of Landowner Actions and Landscape Heterogeneity.” <i>PLoS ONE</i> 20(3): e0320316. https://doi.org/10.1371/journal.pone.0320316. Hoag, Dana, L, Ryan Reuter, Daniel F. Mooney, Jeff Vitale, John Ritten, Nathan D. DeLay, Paul Evangelista, and Anthony Vorster. 2025. “The Economic Fundamentals of Virtual Fencing.” <i>Rangelands</i> 47(1): 92–101. https://doi.org/10.1016/j.rala.2024.05.006 DeLay, Nathan D., Daniel F. Mooney, and John Ritten. 2025. “Climate and Consolidation in the U.S. Beef Cow Sector.” <i>Journal of Agricultural and Resource Economics</i> 50(1): 97–118. https://doi.org/10.22004/ag.econ.344684

Nathan D. DeLay, Daniel F. Mooney, John P. Ritten, and Dana L. Hoag. 2024. "Virtual Fencing: Economic and Policy Implications of an Emerging Livestock Technology." *Western Economics Forum* 22(1): 54–64.

DeLay, Nathan D., James M. Mintert, Nathanael M. Thompson, Jayson L. Lusk, and Todd H. Kuethe. 2024. "Value of Farm Data in Farmland Rental Markets: Management vs. Signaling Value." *Land Economics* 100(4): 690–708.
<https://doi.org/10.3368/le.100.4.041023-0032R1>

Comstock, Haden* and **Nathan D. DeLay**. 2023. "Estimating U.S. Farmers Speed of Climate Change Adaptation: The Case of Subsurface Tile Drainage." *Agricultural Finance Review* 83(4/5): 734–761.
<https://doi.org/10.1108/AFR-02-2023-0027>

DeLay, Nathan D., Michael D. Boehlje, and Shannon Ferrell. 2023. "The Economics of Property Rights in Digital Farming Data: Implications for Farmland Markets." *Applied Economic Perspectives and Policy* 45(4): 2106–2126.
<https://doi.org/10.1002/aepp.13340>

DeLay, Nathan D., Nathanael M. Thompson, and James R. Mintert. 2022. "Precision Agriculture Adoption and Technical Efficiency." *Journal of Agricultural Economics* 73(1): 195–219. (Wiley Top Cited Article 2022–2023)
<https://doi.org/10.1111/1477-9552.12440>

DeLay, Nathan D., Brady E. Brewer, Allen M. Featherstone, and David Boussios. 2022. "The Impact of Crop Insurance on Farm Financial Outcomes." *Applied Economic Perspectives and Policy* 45(1): 579–601.
<https://doi.org/10.1002/aepp.13223>

Thompson, Nathanael M., Megan N. Hughes*, Eugene K.M. Nuworsu*, Carson J. Reeling, Shalamar D. Armstrong, James R. Mintert, Michael R. Langemeier, **Nathan D. DeLay**, and Kenneth. A. Foster. 2022. "Opportunities and Challenges Associated with "Carbon Farming" for U.S. Row-Crop Producers." *Choices* Quarter 3.
<https://doi.org/10.22004/ag.econ.329528>

Chaterji, Somali, **Nathan D. DeLay**, John Evans, Nathan Mosier, Bernard Engel, Dennis Buckmaster, Michael Ladisch, and Ranveer Chandra. 2021. "LATTICE: Machine Learning, Data Engineering, and Policy Considerations for Digital Agriculture at Scale." *IEEE Open Journal of the Computer Society* 2: 227–240.
<https://doi.org/10.1109/OJCS.2021.3085846>

Thompson, Nathanael M., **Nathan D. DeLay**, and James R. Mintert. 2021. "Understanding the Farm Data Lifecycle: Collection, Use, and Impact of Farm Data on U.S. Commercial Corn and Soybean Farms." *Precision Agriculture* 22(6): 1685–1710.
<https://doi.org/10.1007/s11119-021-09807-w>

DeLay, Nathan D., James R. Mintert, and Nathanael M. Thompson. 2021. "Farm Data Collection and Software Adoption in Commercial Scale U.S. Corn-Soybean Farms." *Western Economic Forum* 19(2): 12–19.

DeLay, Nathan D. and Haden Comstock*. 2021. "Recent Trends in PA Technology Adoption and Bundling in Corn Production: Implications for Farm Consolidation." *Western Economic Forum* 19(2): 44–57.

DeLay, Nathan D., Hayley H. Chouinard, Cory G. Walters, and Philip R. Wandschneider. 2020. "The Influence of Crop Insurance Agents on Coverage Choices: The Role of Agent Competition." *Agricultural Economics* 51(4): 623–638.
<https://doi.org/10.1111/agec.12576>

DeLay, Nathan D., S. M. Thumbi, Julia Vanderford, Elkanah Otiang, Linus Ochieng, M. Kariuki Njenga, Guy H. Palmer, and Thomas L. Marsh. 2020. “Linking Calving Intervals to Milk Production and Household Nutrition in Kenya.” *Food Security* 12: 309-325.

<https://doi.org/10.1007/s12571-019-01006-w>

DeLay, Nathan D. 2019. “The Impact of Federal Crop Insurance on the Conservation Reserve Program.” *Agricultural and Resource Economics Review* 48(2): 297-327.

<https://doi.org/10.1017/age.2019.9>

DeLay, Nathan D., Hayley H. Chouinard, and Philip R. Wandschneider. 2019. “Categorizing Agritourism Operations and Identifying Barriers to Success.” *Journal of Agribusiness* 37(1): 1-22.

Under Review

Bista, Bishal*, **Nathan D. DeLay**, Diego Manriquez-Alvarez, and Pablo Pinedo. “Integrating Automated Body Condition Scores and Lactation Data via Optimization Algorithms for Maximized Milk Revenue and Minimized Cost of Delayed Conception in Dairy Cows.” Under review at *Journal of Dairy Science*.

Stanley, Paige, Alexandria S. Kuhl, Jason E Rowntree, Isabella De Faria Maciel, M. Francesca Cotrufo, Douglas J Goodwin, Jennifer Hodbod, Martha C Anderson, John D Scasta, Micaela Branecky, Sarah Brokus, Robert J Clement, **Nathan D DeLay**, Justin D Derner, Morgan MathisonSlee, Rebecca Mitchell, Yao Zhang, Jeremiah Asher, Feng Gao, Dale T Manning, ..., Jonathan Vivas. “Merging Rigor and Relevance in Grazingland Research: A Comprehensive Social-Ecological Monitoring Approach.” Under review at *Carbon Balance and Management*.

Swenson, Maya A.*, Juan J. Vargas, John P. Ritten, Ashley K. Schilling-Hazlett, Edilane C. Martins, Willian A. Souza, Erin N. Burke, Rhyse K. Campion, Anna M. Shadbolt, Pedro H.V. Carvalho, Kimberly R. Stackhouse-Lawson, **Nathan D. DeLay**, Sara E. Place. “Effects of Growth Hormone Implants on Animal Growth Performance, Feed Intake, Carcass Characteristics, Feeding Behavior, Gas Flux Emissions, and Relative Profitability of Finishing Angus Steers.” Under review at *Translational Animal Science*.

Work in Progress

DeLay, Nathan D., Brady E. Brewer, Ty Kreitman, Francisco Scott, and Cortney Cowley. “Fed Cattle and Fed Policy: The Role of Interest Rates in the U.S. Beef Cattle Cycle.” (Submitted for presentation at 2026 AAEA, target journal: *American Journal of Agricultural Economics*).

Bista, Bishal* and **Nathan D. DeLay**. “Willingness to Participate in Methane-Reduction Markets: Insights from U.S. Cattle Producers.” (Submitted for presentation at 2026 AAEA, target journal: *Food Policy*).

Morse, Madison*, Jason Lombard, **Nathan D. DeLay**, Kathy Whitman, and Natalie Urie. “Short Run Direct Financial Impact of HPAI H5N1 on Dairy Farms in the United States.” (Submitted for presentation at 2026 AAEA, target journal: TBD).

Bista, Bishal*, **Nathan D. DeLay**, Sophie McKee, and Daniel Mooney. “Invasive Wildlife and Agricultural Risk: A Spatial Panel Analysis of Feral Pig Damage.” (Submitted for presentation at 2026 AAEA, target journal: *American Journal of Agricultural Economics*).

Manning, Dale, Yining Wu, Frank Lupi, Dabit Bista*, and **Nathan D. DeLay**. “Do Ranchers Avoid Uncertain Conservation Payments? Measuring Risk Preferences for Ecosystem Service Contracts.” (Submitted for presentation at 2026 AAEA, target journal: TBD)

Wi, Yining, Dale Manning, Frank Lupi, Dabit Bista*, and **Nathan D. DeLay**. “Optimizing Grazing Management for Profit and Ecosystem Services: An Integrated Biogeochemical-Economic Model.” (Submitted for presentation at 2026 AAEA, target journal: TBD)

DeLay, Nathan D., Bishal Bista*, and John Ritten. “Simulating Profit and Producer Decision-Making in Methane Reduction Markets for the Fed Cattle Industry.” (Submitted for presentation at 2026 WAEA, target journal: TBD).

de Menezes, Tais*, **Nathan D. DeLay**, and Amanda M. Countryman. “Integrated Modeling of Policy Changes: The Case of Foot-and-Mouth Disease Vaccination in Brazil.” (Presented at 2025 AAEA/WAEA, target journal: *Food Policy*).

Bista, Bishal*, **Nathan D. DeLay**, and Alessandro Bonanno. “Impact of Pasture, Rangeland, and Forage (PRF) Insurance on Reinsurance Fund Allocation.” (Presented at 2025 AAEA/WAEA, target journal: *Applied Economic Perspectives and Policy*).

DeLay, Nathan D. “Charity Hazard on the Range: Forage Disaster Aid and Crowding Out of Pasture Insurance.” (Target journal: TBD).

Grants

USDA NIFA AFRI Foundational and Applied Science
“Combining Earth Observations with Machine Learning to Produce the Methane Emissions-Detection, Analysis, and Resource Management Tool: MethaneDART.”
Role: Co-PI
Funding amount: \$650,000
Duration: 2026-2029
Submitted: Nov 2025
Status: Pending

USDA NIFA AFRI Foundational and Applied Science
“Estimating U.S. CAFO Ammonia Emissions Using Satellite and Ground Observations for Economic, Health, and Policy Analysis.”
Role: Co-PI
Funding amount: \$650,000
Duration: 2026-2029
Submitted: Oct 2025
Status: Pending

Foundation for Food and Agricultural Research (FFAR)
“Metrics, Management, and Monitoring: An Investigation of Pasture and Rangeland Soil Health and Its Drivers.”
Role: Co-PI
Funding amount: \$19,000,000
Duration: 2021-2026
Submitted: Jan 2021
Status: Awarded

USDA APHIS NWRC Cooperative Agreement
“Economics Analysis of Feral Swine Impacts to Agriculture and Natural Resources”
Role: Co-PI
Funding amount: \$200,163
Duration: 2023-2025
Submitted: July 2023
Status: Approved Aug 15, 2023

USDA NRCS Conservation Innovation Grant Classic
“Innovative Technologies to Reduce Beef Industry GHG Emissions.”
Role: PI

Funding amount: \$2,067,293 (\$1,032,839 Federal)
Duration: 2023-2026
Submitted: Oct 2022
Status: Approved April 1, 2023

Purdue AgSEED - Agricultural Research and Extension Leading to Economic Development in Indiana Agriculture and Rural Communities
“Producer Farm Data Valuation, Capitalization, and Sharing.”
Role: Principal Investigator
Funding amount: \$50,000
Duration: 2021-2022
Submitted: Oct 2020
Status: Approved Jan 9, 2021

USDA NIFA Food and Agriculture Cyberinformatics and Tools Initiative (FACT)
“An Innovative Cyber-Framework Integrating Public/Private Data for Evidence-Based Recommendations.”
Role: Co-PI
Funding amount: \$1,000,000
Duration: 2019-2024
Submitted: Oct 2018
Status: Approved May 9, 2019

Purdue/CME Group
“Ag Economy Barometer”
Role: Co-PI
Funding amount: \$284,059
Duration: 2020-2022
Status: Approved

Extension & Outreach Publications

Cory G. Walters and **Nathan D. DeLay**. “Where is Your Crop Insurance Agent? Understanding the Geographical Distribution of Agents.” University of Nebraska Cornhusker Economics, April 2025.
<https://agecon.unl.edu/where-your-crop-insurance-agent-understanding-geographical-distribution-agents/>

DeLay, Nathan D., John P. Ritten, and Daniel Mooney. “Climate Adaptation and Concentration in the U.S. Livestock Sector.” CSU AgNext 2023 Research Report.
<https://agnext.colostate.edu/research-report/>

Ritten, John P. and **Nathan D. DeLay**. “Overview of Carbon Markets.” CSU AgNext Blog, March 15, 2023
<https://agnext.colostate.edu/2023/03/15/overview-of-carbon-markets/>

Ritten, John P. and **Nathan D. DeLay**. “Challenges and Outlook of Carbon Markets in Animal Agriculture.” CSU AgNext Blog, March 15, 2023
<https://agnext.colostate.edu/2023/03/15/challenges-and-outlook-of-carbon-markets-in-animal-agriculture/>

Thompson, Nathanael M., Megan N. Hughes, Eugene K.M. Nuworsu, Carson J. Reeling, Shalamar D. Armstrong, James R. Mintert, Michael R. Langemeier, **Nathan D. DeLay**, and Kenneth A. Foster. “Opportunities And Challenges Associated With Carbon Farming For U.S. Row-Crop Producers.” Purdue Center for Commercial Agriculture Report, June 2021.
<https://ag.purdue.edu/commercialag/home/resource/2021/06/opportunities-and-challenges-associated-with-carbon-farming-for-u-s-row-crop-producers/>

Mintert, James R., **Nathan D. DeLay**, and Michael R. Langemeier. 2022. “Perspec-

tives On Digital Farm Data Ownership.” *Proceedings of the 23rd IFMA Congress*.

Comstock, Haden and **Nathan D. DeLay**. 2022. “The Effects of Government Farm Support Programs on the Adoption of Farm Technology and Sustainable Production Practices.” *Purdue Agricultural Economics Report* PAER-2022-16.

DeLay, Nathan D. “Using Technology to Address Farm Labor Shortages - Effective or Not?” Purdue Center for Food and Agricultural Business Quarterly Review Series, March 24, 2021.
<https://agribusiness.purdue.edu/using-technology-to-address-farm-labor-shortages/>

DeLay, Nathan D., Hayley H. Chouinard, Cory G. Walters, and Philip R. Wandschneider. “Examining the Role of the Crop Insurance Selling Agent.” University of Nebraska Cornhusker Economics, April 2020.
<https://agecon.unl.edu/cornhusker-economics/2020/examining-role-crop-insurance-selling-agent>

DeLay, Nathan D., Nathanael M. Thompson, and James R. Mintert. “Farm Data Usage in Commercial Agriculture.” Purdue Center for Commercial Agriculture Report, Jan 2020.
<https://ag.purdue.edu/commercialag/home/resource/2020/01/farm-data-usage-in-commercial-agriculture/>

DeLay, Nathan D. “Understanding Farm Data Usage.” Purdue Center for Food and Agricultural Business blog post, Oct 4, 2019.
<https://agribusiness.purdue.edu/understanding-farm-data-usage/>

Castro, Luciano and **Nathan D. DeLay**. 2019. “Solinftec: Creating a Foothold in the North American Digital Farming Market.” Purdue Center for Food and Agricultural Business case study.

DeLay, Nathan D. “The Blockchain: A New Form of Trust.” Purdue Center for Food and Agricultural Business Quarterly Review Series, March 15, 2019.
<https://agribusiness.purdue.edu/the-blockchain-a-new-form-of-trust/>

DeLay, Nathan D. “Purdue/CME Barometer Drone Response Analysis.” Purdue Center for Commercial Agriculture Annual Report, Dec 2018.
<https://ag.purdue.edu/commercialag/home/wp-content/uploads/2019/11/2018-CCA-Annual-Report-website-PDF.pdf>

Press & Media

Interview, *Colorado Public Radio KUNC*, “Colorado ranchers say White House beef policy is making a tough job even tougher,” Dec. 10, 2025.
<https://www.cpr.org/2025/12/10/colorado-ranchers-trump-beef-policy/>

Interview, *9News Denver*, “Beef prices: Why theyre going up and how to save money,” July 21, 2025.
<https://www.9news.com/article/money/consumer/steve-on-your-side/beef-prices-why-going-up-save-money/73-2b57a491-653a-44b0-9aa5-9d4857eee35f>

Quoted, *Denver Post*, “Colorado beef prices sky high in advance of Fourth of July,” published July 2, 2025
<https://www.denverpost.com/2025/07/02/colorado-beef-prices-expensive-inflation/>

Quoted, *The Colorado Sun*, “Wrecked rain gauges. Whistleblowers. Million-dollar payouts and manhunts. Then a Colorado crop fraud got really crazy,” published Sept. 8, 2024
<https://coloradosun.com/2024/09/08/patrich-esch-ed-dean-jagers-ranchers-fraud-insurance/>

DeLay, Nathan D. and Amanda Countryman. "The Role of Trade in U.S. Agriculture."
AgNext podcast episode 21, May 7, 2025.
<https://agnext.colostate.edu/podcast/>

DeLay, Nathan D. "Conservation Innovation Grant Update."
AgNext podcast episode 14, Sept. 23, 2024.
<https://agnext.colostate.edu/podcast/>

DeLay, Nathan D. "Farm Data Valuation."
AgNext podcast episode 11, June 12, 2024.
<https://agnext.colostate.edu/podcast/>

DeLay, Nathan D. "On Teaching, Grants, and Tradeoffs."
AgNext podcast episode 5, Dec 13, 2023.
<https://agnext.colostate.edu/podcast/>

DeLay, Nathan D. "The Value of Farm Data."
Purdue Center for Commercial Agriculture Commercial AgCast podcast episode, March 24, 2021.
<https://ag.purdue.edu/commercialag/home/resource/2021/03/the-value-of-farm-data/>

DeLay, Nathan D. "Data and Precision Ag Usage."
Purdue Center for Commercial Agriculture Commercial AgCast podcast episode, June 17, 2020.
<https://ag.purdue.edu/commercialag/home/resource/2020/06/data-and-precision-ag-usage/>

DeLay, Nathan D. "Farm data collection and farmer use analyzed." *The Western Producer*, April 16, 2020.
<https://www.producer.com/crops/farm-data-collection-and-farmer-use-analyzed/>

DeLay, Nathan D. "What Data is Being Collected on the Farm and Who Can You Trust With It?" Interview with Hoosier Ag Today, Nov 12, 2019
<https://www.hoosieragtoday.com/data-collected-farm-can-trust/>

DeLay, Nathan D. "Purdue survey looks at farmer use of ag data software." *Indiana Prairie Farmer*, May 30, 2019.
<https://www.farmprogress.com/management/purdue-survey-looks-farmer-use-ag-data-software>

Teaching Experience

Instructor, LBM 333A Livestock Business Engaged Research: Proposal Development, Spring 2025 (Enrollment: 8).
Department of Agricultural and Resource Economics / Department of Animal Sciences, Colorado State University.

Designer, AREC 408 Agricultural Finance Online course redesign, Fall 2024.
Department of Agricultural and Resource Economics, Colorado State University.

Instructor, AREC 408 Agricultural Finance Online, Spring 2024 (Enrollment: 34).
Department of Agricultural and Resource Economics, Colorado State University.

Instructor, AREC 408 Agricultural Finance, Spring 2025 (Enrollment: 26), Spring 2024 (Enrollment: 20), Spring 2023 (Enrollment: 17).
Department of Agricultural and Resource Economics, Colorado State University.

Co-Instructor, AGR 333 Data Science for Agriculture, Spring 2024 (Enrollment: 21), Spring 2023 (Enrollment: 24), Spring 2022 (Enrollment: 18), Spring 2021 (Enrollment: 14), Fall 2020 (Enrollment: 7).
College of Agriculture, Purdue University.

Instructor, AGECE 684 Applied Quantitative Methods, Fall 2023 (Enrollment: 18), Fall 2022 (Enrollment: 14), Fall 2021 (Enrollment: 42), Fall 2020 (Enrollment: 32), Fall 2019 (Enrollment: 18).

Department of Agricultural Economics MS/MBA Program, Purdue University.

Instructor, AGECE 203 Introductory Microeconomics for Food & Agribusiness, Fall 2021 (Enrollment: 103), Fall 2020 Online (Enrollment: 155), Fall 2019 (Enrollment: 165).

Department of Agricultural Economics, Purdue University.

Guest lecture, AGECE 375 The Process of Economic Research, Spring 2021 (Enrollment: 13).

Department of Agricultural Economics, Purdue University.

Instructor, AGECE 202 Spreadsheet Use in Agricultural Business, Spring 2019 (Enrollment: 92).

Department of Agricultural Economics, Purdue University.

Instructor, EconS 321 Sport Economics Online, Summer 2018 (Enrollment: 18).

School of Economic Sciences, Washington State University.

Masters Level Online Econometrics course design with Thomas L. Marsh, Fall 2017.

School of Economic Sciences, Washington State University.

Instructor, EconS 404 Economics for Managers Online, Summer 2017 (Enrollment: 34).

School of Economic Sciences, Washington State University.

Instructor, EconS 352 Business Management Economics, Spring 2017 (Enrollment: 45), Spring 2016 (Enrollment: 62).

School of Economic Sciences, Washington State University.

Teaching Assistant, EconS 101 Fundamentals of Microeconomics & EconS 102 Fundamentals of Macroeconomics, Fall 2014, Spring 2015.

School of Economic Sciences, Washington State University.

Student Advisement

Advisor. Bishal Bista (Ph.D.). Anticipated graduation: Spring 2027.

Committee Member. Tais de Menezes (Ph.D.). Anticipated graduation: Spring 2026.

Committee Member. Seonhyun Kim (Ph.D.). Anticipated graduation: Spring 2026.

Committee Member. MacKenzie Griffin (M.S. AnSci). Graduated: Fall 2025.

Committee Member. Maya Swenson (M.S. AnSci). Graduated: Spring 2025.

Advisor. Haden Comstock (M.S.). Graduated: Spring 2022. AAEEA Outstanding Thesis 2023.

Co-Advisor. Douglas Abney (M.S.). Graduated: Spring 2022.

Committee Member. Ahmad Wahdat (Ph.D.). Graduated: Spring 2021.

Presentations

Bista, Bishal and Nathan D. DeLay. "Willingness to Participate in Methane Reduction Markets: Insights from U.S. Cattle Producers."

2025 CSU AgNext Industry Innovation Working Group Fall Meeting. Nov 13, 2025. Fort Collins, CO.

de Menezes, Tais, Nathan D. DeLay, Amanda Countryman. “Integrated Modeling of Policy Changes: The Case of Foot-and-Mouth Disease Vaccination in Brazil.” 2025 AAEE/WAEA Annual Meeting. Denver, CO.

Bista, Bishal, Nathan D. DeLay, and Alessandro Bonanno. “Impact of Pasture, Rangeland, and Forage (PRF) Insurance on Reinsurance Fund Allocation.” 2025 AAEE/WAEA Annual Meeting Poster Presentation. Denver, CO.

DeLay, Nathan D. “Innovative Technologies to Reduce Beef Industry GHG Emissions.” 2025 CSU AgNext Research Summit. June 11, 2025. Fort Collins, CO.

Bista, Bishal, Nathan D. DeLay, and Alessandro Bonanno. “Impact of Pasture, Rangeland, and Forage (PRF) Insurance on Reinsurance Fund Allocation.” 2025 CSU AgNext Research Summit Poster Competition. Fort Collins, CO.

Bista, Bishal, Nathan D. DeLay, Diego Manriquez Alvarez, and Pablo Pinedo. “Efficient Dairy Management: Leveraging NSGA-II for Optimized Milk Production and Fertility.” 2024 CSU AgNext Industry Innovation Working Group Fall Meeting. Nov 6, 2024. Fort Collins, CO.

DeLay, Nathan D., Kim Stackhouse-Lawson, Sara Place, John Ritten, Mark Enns, EJ Raynor, Anna Shadbolt, Justin Derner, and Larry Kuehn. “Innovative Technologies to Reduce Beef Industry Greenhouse Gas Emissions.” 2024 SWCS Annual Meeting, NRCS CIG Showcase Poster Presentation. Myrtle Beach, SC.

DeLay, Nathan D., Sophie McKee, and Daniel Mooney. “The Importance of Individual and Collective Action in Wild Pig Damage to U.S. Crop and Livestock Farms.” 2024 WAEA Annual Meeting. San Francisco, CA.

Bista, Bishal, Nathan D. DeLay, Diego Manriquez Alvarez, and Pablo Pinedo. “Efficient Dairy Management: Leveraging NSGA-II for Optimized Milk Production and Fertility.” 2024 WAEA Annual Meeting. San Francisco, CA.

DeLay, Nathan D. “Economics Research: Climate, Policy, and Technology.” 2024 CSU AgNext Industry Innovation Working Group Spring Meeting. May 30, 2024. Fort Collins, CO.

DeLay, Nathan D., Daniel Mooney, and John P. Ritten. “Climate and Consolidation in the U.S. Beef Cattle Industry.” CSU Animal Sciences General Seminar. Feb 28, 2024. Fort Collins, CO.

DeLay, Nathan D., Daniel Mooney, and John P. Ritten. “Climate and Consolidation in the U.S. Beef Cattle Industry.” 2024 International Livestock Forum. Jan 8, 2024. Denver, CO.

DeLay, Nathan D. “Capitalizing Farm Data for Sustainable Agriculture.” 2023 Soil-based Climate Solutions: Systems-level Transformations to Enable Soil-based Climate Solutions at Scale. Sept 29, 2023. Denver, CO.

DeLay, Nathan D. “AgNext Research and Outreach in Collaboration with Extension.” 2023 CSU Office of Engagement and Extension Fall Forum. Sept 20, 2023. Fort Collins, CO.

DeLay, Nathan D., Daniel Mooney, and John P. Ritten. “Climate Adaptation and Concentration in the U.S. Livestock Sector.” 2023 WAEA/CAES Annual Meeting. Whistler, B.C.

Mooney, Daniel and Nathan D. DeLay. “Not So Flat Payoff Functions? Production

Risk and Economically Optimal Irrigation Decisions.”
2023 WAEA/CAES Annual Meeting. Whistler, B.C.

Walters, Cory and Nathan D. DeLay. “The Influence of the 2011 Standard Reinsurance Agreement Cap on Crop Insurance Agent Commissions. Did Agents Expand or Leave?”
2023 WAEA/CAES Annual Meeting. Whistler, B.C.

DeLay, Nathan D. “Technology and Rural Communities.”
2023 CSU AgNext Research Summit. June 14, 2023. Fort Collins, CO.

DeLay, Nathan D. “AgNext Agricultural Economics Research and Teaching Priorities.”
2023 CSU AgNext Industry Innovation Working Group Spring Meeting. June 1, 2023. Fort Collins, CO.

DeLay, Nathan D. “Opportunities and Challenges with Carbon Farming.”
2022 Soil-based Climate Solutions: Science to Drive Impact conference. Nov 12, 2022. Fort Collins, CO.

Comstock, Haden and Nathan D. DeLay. “The Effects of Government Farm Support Programs on the Adoption of Farm Technology and Sustainable Production Practices.”
2022 AAEA Annual Meeting Poster Presentation. Anaheim, CA.

Minert, James R., Nathan D. DeLay, and Michael R. Langemeier. “Perspectives On Digital Farm Data Ownership.”
23rd International Farm Managers Association Congress 2022. Copenhagen, Denmark.

DeLay, Nathan D. “Innovations in Farm Technology & Data”
2022 Commodity Classic Conference. New Orleans, LA.

DeLay, Nathan D., James R. Mintert, Brady Brewer. “Data and Technology: Insights and Implications for Serving the Farm of the Future .”
2021 Purdue Center for Food and Agricultural Business National Conference. West Lafayette, IN.

DeLay, Nathan D. and Cory G. Walters. “Agent Location in the Federal Crop Insurance Program: A Spatial Regression Analysis.”
2021 AAEA Annual Meeting. Austin, TX.

DeLay, Nathan D., Nathanael M. Thompson, and James R. Mintert. “Farm Data Usage and Analytics.”
2021 AAEA Annual Meeting. Austin, TX.

DeLay, Nathan D. “Putting a Price on Data.”
Purdue College of Agriculture Spring Webinar Series: Data Driven Agriculture, April 1, 2021.
<https://ag.purdue.edu/digital-ag-resources/event/webinar-putting-a-price-on-data/>

DeLay, Nathan D., James R. Mintert, Nathanael M. Thompson. “Growing Your Digital Agriculture Strategy - Evidence from Purdue Research.”
Purdue Center for Commercial Agriculture Webinar, Sept 16, 2020.
<https://ag.purdue.edu/commercialag/home/resource/2020/09/growing-your-digital-agriculture-strategy/>

DeLay, Nathan D. and Scott Downey. “Key Findings from Recent Research over Farmers Use of Data.”
Purdue Center for Food and Agricultural Business Tuesday Tailgate webinar, June 2, 2020.
<https://agribusiness.purdue.edu/tuesday-tailgate-talks/archive/>

DeLay, Nathan D., Nathanael M. Thompson, and James R. Mintert. "Precision Agriculture Adoption and Technical Efficiency."
2020 AAEA Annual Meeting (virtual). Kansas City, Missouri.

DeLay, Nathan D. "The Promise of Digital Agriculture: Insights from Crop Farmer Digital Ag Technology Usage Survey."
2020 Purdue Center for Commercial Agriculture Advisory Council Meeting. West Lafayette, IN.

DeLay, Nathan D. "Growing Your Digital Agriculture Strategy."
2020 Purdue Top Farmer Conference. West Lafayette, IN.

DeLay, Nathan D. "Farm Data Use in Commercial Agriculture."
2019 Purdue Center for Food and Agricultural Business National Conference. West Lafayette, IN.

DeLay, Nathan D. "Blockchain for Agriculture: Promise, Potential, and Reality."
2019 Purdue Center for Food and Agricultural Business Executive Summit. West Lafayette, IN.

DeLay, Nathan D. "Blockchain for Agriculture: Promise, Potential, and Reality."
2019 Purdue University Digital Agriculture Roundtable. West Lafayette, IN.

DeLay, Nathan D., Brady E. Brewer, Allen M. Featherstone, and David Boussios. "The Impact of Crop Insurance on Farm Financial Outcomes."
2019 AAEA Annual Meeting. Atlanta, GA.

DeLay, Nathan D., Brady E. Brewer, Allen M. Featherstone, and David Boussios. "The Impact of Crop Insurance on Farm Financial Outcomes."
2019 WAEA Annual Meeting. Coeur d'Alene, ID.

DeLay, Nathan D. "Drone Use in Agriculture: How Common are They?"
Spring 2019 IN Farm Bureau/Purdue Ag Econ Supper Series. Indianapolis, IN.

Panel Moderator, Using Technology Beyond Precision Applications to Improve Profitability.
2019 Purdue Top Farmer Conference. West Lafayette, IN.

Participant, James M. Buchanan's Legacy for Liberty Conference.
2018 IHS/Liberty Fund Advanced Topics in Liberty Conference Series. Arlington, VA.

DeLay, Nathan D., Hayley H. Chouinard, Cory G. Walters, and Philip P. Wandschneider. "The Market Structure for Crop Insurance and the Effects on Insurance Contracts."
2017 School of Economic Sciences Seminar Series. Pullman, WA.

DeLay, Nathan D., Hayley H. Chouinard, Cory G. Walters, and Philip P. Wandschneider. "The Market Structure for Crop Insurance and the Effects on Insurance Contracts."
2017 AAEA Annual Meeting Poster Presentation. Chicago, IL.

DeLay, Nathan D. "The Impact of Federal Crop Insurance on the Conservation Reserve Program."
2017 School of Economic Sciences GSA Poster Expo. Pullman, WA.

DeLay, Nathan D. "The Impact of Federal Crop Insurance on the Conservation Reserve Program."
2016 Property and Environment Research Center (PERC) Summer Workshop Series. Bozeman, MT.

DeLay, Nathan D., Hayley H. Chouinard, and Philip P. Wandschneider. "Categorizing Agritourism Operations and Identifying Barriers to Success." 2016 WAEA/CAES Annual Meeting. Victoria, B.C.

Fellowships & Awards

2023, AAEA Outstanding Master's Thesis advisor. Washington, D.C.

2021, Outstanding M.S./M.B.A. Capstone Advisor Award. West Lafayette, IN.

2020, Acorn Award, Purdue Seed for Success Research. West Lafayette, IN.

Purdue College of Agriculture Elevating the Visibility of Agricultural Research: 150th Anniversary Review Papers Award, 2019. West Lafayette, IN.

School of Economic Sciences Excellence in Teaching Award, Fall 2017. Pullman, WA.

Institute for Humane Studies at George Mason University Ph.D. Scholarship, Fall 2017–Spring 2018. Arlington, VA.

School of Economic Sciences GSA Poster Competition Honorable Mention, Spring 2017. Pullman, WA.

Property and Environment Research Center (PERC) Graduate Fellowship, Summer 2016. Bozeman, MT.

2016 WAEA/CAES Annual Meeting Travel Grant Recipient, Summer 2016. Victoria, B.C.

School of Economic Sciences Tesfaye Girma Deboch Graduate Fellowship, Summer 2016. Pullman, WA.

Rocky Mountain College Full Tuition Scholarship, Fall 2005–Spring 2009. Billings, MT.

Professional Affiliations

Agricultural and Applied Economics Association (AAEA)

Western Agricultural Economics Association (WAEA)

Professional Service

Referee, *Food Policy*, *American Journal of Agricultural Economics*, *Applied Economic Perspectives and Policy*, *Journal of Agricultural Economics*, *Journal of Agricultural and Resource Economics*, *Journal of the Agricultural and Applied Economics Association*, *Agricultural Economics*, *Agricultural and Resource Economics Review*, *Journal of Agricultural and Applied Economics*, *Agricultural Finance Review*, *Choices*, *Western Economic Forum*, *Agronomy Journal*.

Director, Western Agricultural Economics Association (WAEA), 2025-2028.

Faculty Advisor, CSU Livestock Business Management (LBM) Club, 2025-current.

Judge, Graduate Student Paper Competition, Western Agricultural Economics Association (WAEA) Annual Meeting, 2024.

Associate Editor, *Agronomy Journal*, Jan. 2022-Jan. 2023.

Judge, Colorado FFA Business Marketing Competition, 2023.

Council Member, WAEA Executive Council, 2023-current.

Committee Member, CSU DARE Undergraduate Curriculum Committee, 2022-current.

Committee Member, Purdue/IU MS/MBA Admissions Committee, 2021-2022.

Reviewer, Purdue/IU MS/MBA Founder's Cup Award for best capstone project, 2021.

Reviewer, Best Paper Award, AAEA Applied Risk Analysis (ARA) section, 2021.

Reviewer, Purdue Department of Agricultural Economics Best Masters Thesis Award, 2021, 2022.

Reviewer, Selected Papers, Agricultural and Applied Economics Association (AAEA) Annual Meeting, 2018, 2019, 2023, 2025.

Reviewer, Selected Papers, Western Agricultural Economics Association (WAEA) Annual Meeting, 2020, 2023, 2024, 2025.

Faculty Advisor, Purdue Data Mine undergraduate and graduate team, 2020-2022.

Team Advisor, Purdue Academic Quiz Bowl Team, 2020-2022.

Faculty advisor, Purdue College of Agriculture Research Data Ecosystem Initiative, Nov 2019-2022.

Team Advisor, 2019 AAEA Graduate Student Case Study Competition.

Grader, Purdue Department of Agricultural Economics graduate student preliminary examinations, 2019, 2021.

Judge, 2019 Department of Agricultural Economics Graduate Student Poster Competition.

Reviewer, 2019 Purdue Research Foundation (PRF) Research Grant Competition.

Reviewer, 2019 Purdue College of Agriculture AgSEED Grant Competition.

Session Moderator, Selected Paper Session 11: Market Structures and Supply Chains. 2016 WAEA/CAES Annual Meeting. Victoria, B.C.